

AARAV RAJ SHRESTHA

Full-Stack Developer — AI Systems — RAG, Real-Time Applications, Scalable Backends

Kathmandu, Nepal

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EDUCATION

SRM University AP

2023 – 2027

B.Tech in Computer Science and Engineering | CGPA: 8.6/10

Andhra Pradesh, India

– **Core Courses:** Data Structures and Algorithms, Operating Systems, DBMS, OOP with C++, Computer Networks

Modern Indian School

2020 – 2023

Higher Secondary Education – PCM

Kathmandu, Nepal

EXPERIENCE

CodingMountain

Jun 2025 – Aug 2025

Software Development Intern

Kathmandu, Nepal

- Designed and developed a scalable real estate marketplace using the PERN stack, handling authentication, listings, and search workflows.
- Improved backend performance by **32%** through API restructuring and database query optimization.
- Enhanced user engagement by implementing secure Google OAuth authentication and nested comment systems.

PROJECTS

RAG-Based PDF Question Answering System (Key Project) — FastAPI, Qdrant, Gemini API [\[GitHub\]](#)

- Designed and built a Retrieval-Augmented Generation (RAG) system for semantic search over PDFs, optimizing retrieval accuracy vs latency using vector embeddings.
- Integrated Qdrant with Gemini LLM to generate context-aware, source-grounded answers, reducing hallucinations.

HabitFlow — React, Node.js, Express.js, MongoDB, JWT, Google OAuth [\[GitHub\]](#)

- Designed and built a full-stack habit tracking system with secure OAuth 2.0 + JWT authentication and REST APIs.
- Implemented habit tracking, streaks, and analytics dashboards with heatmaps to visualize user behavior.
- Optimized backend performance, reducing habit fetch latency by 42% through query and API improvements.

Real-Time Chat Application — MERN, Socket.IO [\[GitHub\]](#)

- Designed and built a real-time chat system using Socket.IO, enabling low-latency bidirectional communication.
- Implemented Google Sign-In, REST APIs, and a reliable notification system for seamless user interaction.
- Structured a monorepo with separate Express backend and React/Vite frontend to improve scalability and maintainability.

Smart Home Security System — React, Node.js, Tailwind, Python, OpenCV, CNN [\[GitHub\]](#)

- Designed a CNN-based facial recognition system achieving **96.8% accuracy** with real-time detection under **200ms**.
- Built a real-time alert pipeline and integrated package detection to enhance security monitoring.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Java, SQL, PHP

Technologies/Frameworks: React.js, Node.js, Express.js, MongoDB, PostgreSQL, HTML5, CSS3, Bootstrap, Tailwind CSS

ML / AI Tools: NumPy, Pandas, Matplotlib, scikit-learn, OpenCV, LangChain

Development Tools: Git, GitHub, VS Code, Vercel, Canva

Familiar With: Kafka, Inngest, AWS, Docker

ACHIEVEMENTS

- Solved **300+** coding problems across LeetCode, HackerRank, CodeChef, and GeeksforGeeks.
- Secured **Top 3** position in MSc Hackathon by building a real-time student networking platform using MERN and Socket.IO.

CERTIFICATIONS

- Generative AI Professional — Oracle
- AI Foundations Associate — Oracle